



Continuum modelling of pedestrian flows – Part 2: Sensitivity analysis featuring crowd movement phenomena

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HIGHLIGHTS

- We present a sensitivity analysis of the continuum model.
- The model's capabilities to predict crowd movement phenomena are studied.
- Only under very specific conditions all phenomena can be reproduced.
- The local route choice component is essential to predict dispersion.
- Self-organization arises as a result of the local route choice.

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ABSTRACT

In recent years numerous pedestrian simulation tools have been developed that can support crowd managers and government officials in their tasks. New technologies to monitor pedestrian flows are in dire need of models that allow for rapid state-estimation. Many contemporary pedestrian simulation tools model the movements of pedestrians at a microscopic level, which does not provide an exact solution. Macroscopic models capture the fundamental characteristics of the traffic state at a more aggregate level, and generally have a closed form solution which is necessary for rapid state estimation for traffic management purposes. This contribution presents a next step in the calibration and validation of the macroscopic continuum model detailed in Hoogendoorn et al. (2014). The influence of global and local route choice on the development of crowd movement phenomena, such as dissipation, lane-formation and stripe-formation, is studied. This study shows that most self-organization phenomena and behavioural trends only develop under very specific conditions, and as such can only be simulated using specific parameter sets. Moreover, all crowd movement phenomena can be reproduced by means of the continuum model using one parameter set. This study concludes that the incorporation of local route choice behaviour and the balancing of the aptitude of pedestrians with respect to their own class and other classes are both essential in the correct prediction of crowd movement dynamics.

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1. Introduction

The fatally and severely injured pedestrians of the New Years stampede in Shanghai (2015) demonstrate that docile large-scale crowd movements during music, sports and religious events can develop into highly dangerous unpredictable

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crowd crushes at a moments notice. The newspaper headlines illustrate that managing pedestrian crowd movements is, due to the complex nature of these movements, a very intricate and complex task.

In recent years numerous pedestrian simulation tools have been developed that can support crowd managers and government officials in their tasks. Many of these tools simulate the movements of pedestrians at a microscopic level. These microscopic models determine the speed and direction of each pedestrian as a direct result of the characteristics of the pedestrian and its interactions with its environment. As a result of the high level of detail, microscopic simulation models tend to have long computation times. But more importantly, since these models predict the traffic state as a summation of the actions of individual particles, no exact solution exists that is necessary for management purposes.

Next to microscopic models, also macroscopic models exist. However, up to this moment only a few macroscopic simulation models have been presented. This type of model captures the fundamental characteristics of the traffic state at a more aggregate level. Consequently, a closed form description of crowd movement behaviour can be achieved.

New technologies to monitor pedestrian flows are in dire need of models that allow for rapid state-estimation. Consequently, these new technologies can potentially benefit from the core properties of macroscopic models. One promising macroscopic modelling attempt to be used for such endeavours has been detailed in Ref. [1].

This contribution presents a next step in the development of this continuum model. In this paper represents the second step in the calibration process of a pedestrian simulation model. That is to say, a comprehensive sensitivity analysis of the continuum model is presented. The influence of global and local route choice parameters on the development of crowd movement phenomena, such as dissipation, lane-formation and stripe-formation, is studied. The main contribution of the presented work is an improved understanding of the influence of the density gradient on the development of crowd movement phenomena in the proposed continuum model.

The paper starts with a brief overview of crowd movement phenomena, macroscopic modelling approaches and contemporary calibration attempts (Section 2). At the end of this section a framework for the calibration and validation of pedestrian simulation models is briefly introduced which will serve as point of departure during the remainder of this paper. A description of the continuum model is provided in Section 3. Subsequently, Section 4 details the sensitivity analysis methodology and the case studies. The simulation results, used in this research to assess the sensitivity of the continuum model, are presented and discussed in Section 5. The last section reviews the key findings of this contribution and presents some directions of future research.

2. Background

In this section, the key studies that feature attempts to model pedestrian movements in a continuous fashion and calibrate pedestrian simulation models are presented, which jointly form the foundation of this research. The main types of macroscopic modelling approaches are discussed in Section 2.1. Accordingly, Section 2.2 presents a brief overview recent attempts to ascertain whether pedestrian simulation models predict crowd movement dynamics accurately. Given that only limited calibration endeavours have been undertaken, Section 2.2 does not distinguish between microscopic and macroscopic modelling attempts.

2.1. Macroscopic modelling of operational walking behaviour

While new microscopic modelling approaches are presented frequently, the development of macroscopic pedestrian simulation models has been slow and sporadic. Nevertheless, several instances of macroscopic models have been presented in the last decade, such as network models, hybrid models and continuum models.

The first type of macroscopic simulation models, i.e. network models, makes use of graph theory to simulate the movements of pedestrians through a network. Examples of this type of models are Chalmer et al. [2], Lovas [3] and Daamen [4]. Due to the typical simplistic description of the network and/or the movement dynamics of the pedestrians, many network models use mathematical procedures borrowed from operational research to determine optimal management strategies swiftly. However, due to the lack of detail in the description of the movement dynamics these models are not capable of predicting the effect key features of crowd movements, such as for instance phase transitions and self-organization patterns.

A second type of macroscopic models, i.e. hybrid models, attempts to tap into the benefits of both microscopic and macroscopic simulation models. The microscopic and macroscopic part of a hybrid model are either entirely overlapping in time and space and used sequentially, or used separately at different time intervals or at different areas of the infrastructure depending on the stability of the traffic state. Banerjee et al. [5] and Xiong et al. [6] are examples of this type of macroscopic modelling. In complicated flow situations (i.e. intersecting flow situations) most hybrid models tend to fall back on the microscopic part of the simulation model. Consequently, the computation speed of hybrid models tends to decrease especially at the instances where the complexity and scale of the flow situation renders microscopic modelling impractical.

Models of the last type, continuum models, attempt to describe pedestrian movements as continuous flows in space and time. The inspiration of this type of aggregate simulation models, which feature the movements of large crowds without detailing the agent-based dynamics, has often been found in nature. Hughes [7] was the first to describe the analogy between pedestrian flows and soil mechanics, and used a potential field approach. Several other studies, Treuille et al. [8] and Xiong et al. [9], have further developed these ideas. Several researchers have found other sources of inspiration; such as for instance

gas dynamics (e.g. Ref. [10]), granular flows (e.g. Ref. [11]), animal societies (e.g. Ref. [12]), traffic flows (e.g. Ref. [13]) and microscopic pedestrian simulation models (e.g. Refs. [14,1]).

Many of the models which belong to this last category can only simulate uni-directional flow situations and cannot handle the complexity of multi-class modelling [15]. As a consequence, even though these models are efficient and fast, they can generally not be used to assess large crowd movements within a complex infrastructure (i.e. train station, festival terrain, large square). The multi-class model proposed by Hoogendoorn et al. [1] is one of the few exceptions, since the incorporation of multiple classes allows this model to simulate the interactions between distinct flows of pedestrians. Yet, up to this moment, it is undetermined whether the crowd movements by this model resemble reality.

2.2. Research attempts to ascertain the reliability and validity of pedestrian simulation models

Even though calibration and validation are considered to be essential to determine the reliability and validity of simulation models, researchers currently apply inconsistent procedures or only partially test the simulation tools due to the lack of international standards for verification and validation of pedestrian flow and crowd dynamic simulation tools for general use [16]. Campanella et al. [17] and Kretz [18] mention that most reports on the calibration of pedestrian simulation models focus on a few aggregate aspects of the flow and generally do not show results for different types of flow situations. Campanella et al. [17] also indicate that there is not much evidence that pedestrian simulation models have been calibrated to correctly predict self-organization phenomena. Furthermore, the influence of the behavioural parameters of pedestrian simulation models on the predicted movement dynamics is scarcely analysed [19].

Recently more studies have been detailing attempts to determine which calibration technique to use (e.g. Refs. [19,20]), assess the sensitivity of model parameters (e.g. Ref. [21]), calibrate models (e.g. Ref. [22]), or validate models (e.g. Ref. [23]). As one can see, these studies mention very distinctive forms of calibration. Even though, the studies have touched upon essential parts of the calibration and validation process.

The authors have attempted to put the methodological procedures mentioned in these studies into perspective in Fig. 1. The figure describes the entire process which a pedestrian simulation model has to go through from initial idea to package for commercial use. The process can be split into four main parts, namely model proposition, calibration, validation and implementation for commercial use. Each of these parts can be divided into several distinctive steps. While the first (1a–1c) and last steps (3a–4) predominantly focus on comprehensively describing the model and its future use, during the intermediate steps (2a–2c) an effort is made to test the capabilities of the model.

Only one step within the calibration process is directed at finding the optimal parameter set of a model. Given the complex behaviour of pedestrian simulation models, calibrating a model's parameters without understanding its sensitivity with respect to the predicted dynamics does not ensure understanding. Consequently, one first needs to assure that the computational procedures are correctly implemented (2a) and that the model can predict realistic behaviour (2b). The first can be tested by means of simple scenario testing based on the theoretical foundation of the model. The latter requires a thorough analysis that determines the sensitivity of a pedestrian simulation model by means of sophisticated scenarios, i.e. scenarios that test the entire range of movement behaviour the model should capture given its future use, and puts the results into perspective by comparing the predicted movement behaviour with the movement behaviour found in real-life situations.

With respect to the development and testing of the model proposed by Hoogendoorn et al. [1], the authors follow the generic framework proposed above. This contribution presents a next step (2a) in the calibration and validation process of the macroscopic continuum model. In the remainder of this paper, the sensitivity of the prediction of crowd movement phenomena with respect to the behavioural parameters of the model is assessed.

3. Description of the continuum model

The continuum model presented in Ref. [1] describes the dynamics of the class-specific density $\rho_d(t, \vec{x})$ over time t and space $\vec{x}$, where d denotes the pedestrian class. In this model, a class describes all pedestrians which walk from the same origin to the same destination, and have the same physical capabilities. The model assumes that the movement behaviour of all pedestrians of a class develops in a similar manner.

3.1. Conservation of pedestrians

Within the continuum model the conservation of pedestrians is assumed. The class-specific conservation of mass equation (Eq. (1)) describes how the density of pedestrians of class d changes over time due to the lateral and longitudinal in- and outflow of pedestrians. In the conservation equation $\vec{q}_d$ represents a two-dimensional flow vector, r_d the inflow of pedestrians and s_d the outflow of pedestrians. The continuum model assumes that conservation holds for every cell defined within the infrastructure.

$$\frac{\partial \rho_d}{\partial t} + \nabla \cdot \vec{q}_d = \vec{r}_d - \vec{s}_d. \quad (1)$$

The equilibrium velocity $\vec{v}_d$ is modelled as a two-dimensional vector that is defined by a speed u_d and a direction of movement $\vec{y}_d$ (Eq. (2)). It is assumed that the speed u_d is a function of the class-specific densities ρ_d (Eq. (3)). As a result, the

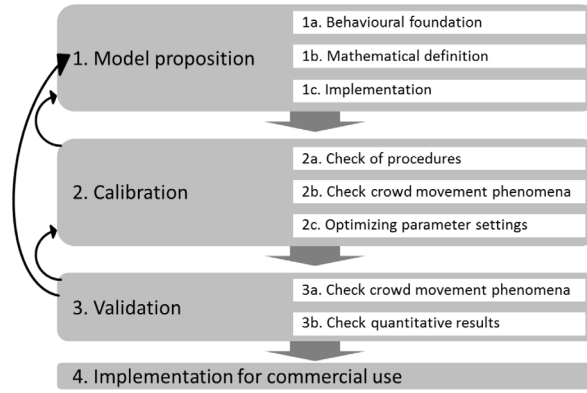


Fig. 1. Generic framework for the proposition, calibration and validation of pedestrian simulation models.

continuum model is capable of simulating the impact of flow composition on the average walking speed for a specific class.

$$\vec{v}_d = u_d \cdot \vec{\gamma}_d \quad (2)$$

$$u_d = U(\rho_1, \dots, \rho_d). \quad (3)$$

The walking direction $\vec{\gamma}_d$ represents the combination of the global and local route choice behaviour of pedestrians. In this study, the global route choice is determined similar to Hoogendoorn and Bovy [24]. That is to say, by means of a value function that represents the minimum cost a class of pedestrians experiences while moving towards a destination, the optimum direction ϕ_d leading to the destination is computed (Eq. (4)). In this case the value function balances the distance with respect to the destination with the disutility of being near static objects. One is referred to Hoogendoorn and Bovy [24] for more details on the computation of the global route choice component.

$$\vec{\gamma}_{global} = -\frac{\nabla \phi_d}{\|\nabla \phi_d\|}. \quad (4)$$

As one can see, the global route choice is not dependent on the traffic state. Consequently, if only taking the global route choice into account, all pedestrians follow their shortest path and will rather wait in line than walk around congestion along their path. To remedy this issue, a local route choice component is introduced. It is assumed that the local route choice can be expressed as a function of the class specific density gradients. This results in the following expression for the flow vector.

$$\vec{q}_d = \vec{\gamma}_d(\rho_1, \dots, \rho_d, \nabla \rho_1, \dots, \nabla \rho_D) \cdot \rho_d \cdot U(\rho_1, \dots, \rho_d). \quad (5)$$

Eq. (5) describes the impact of spatially distinct flow conditions on the local route choice, and hence the flow direction. Pedestrians are assumed to avoid high density areas, in particular when these areas are made up of pedestrians from another class (Eq. (6)). β_d can be interpreted as weights that a pedestrian of group d attaches to densities due to the presence of pedestrians of its own ($\delta = d$) and other classes ($\delta \neq d$).

$$\vec{\gamma}_{local} = -\frac{\beta_{\delta=d} \cdot \nabla \rho_{\delta=d} + \beta_{\delta \neq d} \cdot \nabla (\sum \rho_{\delta \neq d})}{\|\beta_{\delta=d} \cdot \nabla \rho_{\delta=d} + \beta_{\delta \neq d} \cdot \nabla (\sum \rho_{\delta \neq d})\|}. \quad (6)$$

3.2. Combining global and local route choice

This leads to the following route choice model:

$$\vec{\gamma}_d = -\frac{\beta_g \cdot \vec{\gamma}_{global} + \beta_l \cdot \vec{\gamma}_{local}}{\|\beta_g \cdot \vec{\gamma}_{global} + \beta_l \cdot \vec{\gamma}_{local}\|} \quad (7)$$

where β_g and β_l represent the influence of the global and local route choice components on the final route choice. Dissimilar to Hoogendoorn et al. [1], the global and local components of the route choice are first transformed into unit-vectors before addition. As a consequence, the weight factors always have the same effect on the pedestrian flow, irrespective of the geometry and size of the infrastructure. Furthermore, the interaction with obstacles, such as walls, will also be standardized.

4. Case studies

In this section the case studies used in the sensitivity analysis are introduced. Firstly, the characteristics of the crowd movement phenomena, often mentioned in the research literature, are reviewed in Section 4.1. In order to predict these phenomena several flow situations, and hence the walking infrastructure, are designed in Section 4.2. Accordingly, the behavioural characteristics are mentioned in Section 4.3. The last part of this section specifies the studied parameter settings 4.4 and the indicators used in the sensitivity analysis 4.5.

4.1. Empirical evidence of crowd movement phenomena

A number of phenomena have been mentioned in the research literature which arise during crowd movements. The most well-known phenomena are the self-organization patterns described by Helbing and Molnar [25], Helbing and Molnar [26] and Hoogendoorn and Daamen [27], namely lane formation, stop-and-go waves, and turbulence. The first of these patterns details the separation of pedestrians in a bi-directional flow situation into several uni-directional streams within the same corridor [28]. The second pattern is characterized by temporarily interrupted longitudinally unstable flows. The third pattern constitutes of random, unintended displacements of pedestrians into all possible directions due to the development of force networks in the crowd [29]. While the latter two patterns occur only in fairly dense crowds, lane formation also arises in low and medium dense crowds. Besides these three patterns, the formation of stripes in intersecting flows is mentioned in the literature (first depicted by Ando et al. [30]—in Japanese).

In addition to the self-organization patterns, there are also phenomena which only arise within specific types of infrastructure, for example the tendency of pedestrians to keep a minimum distance with respect to walls overview of studies presented in Ref. [31]. Another trend mentioned in the literature is the fact that pedestrians tend to move towards the inner wall just upstream of the bend and shy away from the wall downstream of a bend (depicted in Ref. [32]). While during bottleneck situations dispersion [33] and anticipation [34] are found. Where the first details the anticipation of pedestrians on the traffic state ahead of them, the latter details the funnel-shaped outflow of pedestrians downstream of a bottleneck.

As mentioned before, the aim of this study is to use a macroscopic simulation model to predict all kinds of intricate crowd movements. This is considered to be only possible if the model can realistically predict most crowd movement phenomena mentioned in the research literature which occur during general conditions. This study tests whether the continuum model is indeed capable of predicting the mentioned phenomena.

4.2. Flow situations and infrastructure lay-out

The previous paragraphs have mentioned a number of crowd movement phenomena which develop as a result of specialistic flow situations. These specific flow situations, i.e. the four situations mentioned in Section 4.1, need to be studied in order to assess the sensitivity of the continuum model. Therefore, in the remainder of this study four distinct flow situations are distinguished:

- *Uni-directional—Short bottleneck*, where one class of pedestrians is generated on the left, walks through a bottleneck of limited size (4 m) and exits on the right. This scenario assesses the development of a funnel-shaped set of trajectories and depicts to which extent anticipation, flow concentration and flow dispersion occurs.
- *Uni-directional—Corner*, where one class of pedestrians is generated on the left and exits at the bottom after making a sharp 90° turn. This scenario illustrates the influence of local route choice on the traffic state at the upstream inside of the corner.
- *Bi-directional—Straight*, where two classes of pedestrians are generated, one from left to right and one from right to left. This scenario provides insights into the capabilities of the continuum model to predict lane-formation and grid-lock.
- *Intersecting flow scenario—90 degrees*, where two classes of pedestrians are generated, one from left to right and one from top to bottom. This scenario produces insights into the occurrence of stripe-formation, turbulence and grid-lock.

In Fig. 2 the lay-outs are depicted which are used to assess the sensitivity of the continuum model. In the case studies rectangular cells of $0.1 \text{ m} \times 0.1 \text{ m}$ are used. The size of the cells determines the coarseness of the computations. There has been specifically chosen for a small cell size in order to see whether the continuum model is capable of estimating crowd dynamics at the same level of detail as a microscopic simulation model. However, a larger the cell size would increase the computation speed of the model. The effect of the cell size is not studied in this paper.

4.3. Pedestrian characteristics

The continuum model uses the fundamental diagram to estimate the impact of density ρ on the decrease of the walking speed u_d . In this study a linear speed–density relation (Eq. (8)) is assumed to test the parameter settings of the continuum model.

$$U(\rho_d) = v_0 \cdot \left(1 - \frac{\rho_d}{\rho_{djam}}\right) = 1.34 \cdot \left(1 - \frac{\rho_d}{5.4}\right). \quad (8)$$

This relation has been chosen for its simplicity and a reduction of computational effort, being fully aware that other, probably more realistic relations, have been put forward in the literature by Weidmann [35].

The sensitivity with respect to the magnitude of the demand is tested using three distinct scenarios, i.e. 1. $q_d = 0.5 \text{ P/m/s}$, 2. $q_d = 1 \text{ P/m/s}$ and 3. $q_d = 2 \text{ P/m/s}$. In order to limit the degrees of freedom in this study, the demand between two flows is always equally divided in case of multiple flows within one infrastructure. As one can see, only light and medium demand levels have been included in this sensitivity analysis. First and foremost because the authors wanted to see how the model handles the phase transitions between no and the existence of self-organization patterns. Therefore, especially

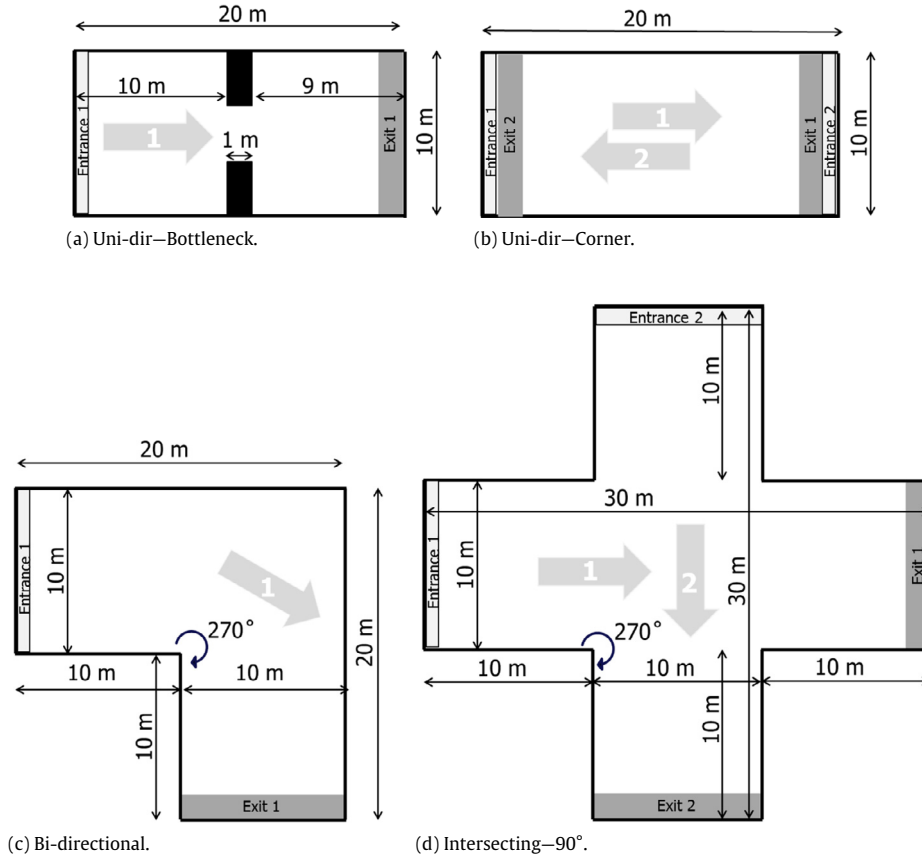


Fig. 2. Infrastructure lay-outs of the simulation case studies.

demand scenarios are included at which these transitions are expected to happen. Besides that, other studies have shown that physical interaction forces between pedestrians can only with great difficulty be handled by the microscopic model which is at the basis of this continuum model. Therefore, it is expected that force networks cannot be modelled by this continuum model.

4.4. Model parameters

The mathematical description of the continuum model mentions four parameters which influence the flow of pedestrians (Table 1). These parameters appear in two pairs. While β_g and β_l determine the relative weight of the global and local route choice components, $\beta_{\delta=d}$ and $\beta_{\delta \neq d}$ determine the balance between the influence of the density gradient of the own and other pedestrian classes in the local route choice components. Given that all four weights are imposed on unit-vectors, only the relative weight of these β 's is accounted for. Furthermore, the authors assume that pedestrians display goal-oriented behaviour. That is to say, β_g is assumed to be always larger or equal to β_l in order to attain movement towards their destination. Moreover, the influence of the density gradient of the other class is assumed to be larger or equal to the density gradient of the own pedestrian class. Last of all, because pedestrians have the tendency to follow others who walk in a similar direction, the authors also assume that they are more attracted towards their own class than towards another class of pedestrians. Taking these assumptions into account, this study tests the sensitivity of the predictions with respect to the parameter settings mentioned in Table 2.

4.5. Indicators of performance

In this study the traffic state predicted by the continuum model is taken as an indicator of the sensitivity of the model. The spatial distribution of the density and velocity serve as a proxy for the traffic state. Additionally, in order to quantify the dispersion of pedestrians at several key cross-sections within the infrastructure, the traffic state has been translated into distributions of density at a certain cross-section. Moreover, the dominant class of pedestrians within each cell has been determined in order to visualize the development of self-organization patterns.

Table 1

Model parameters and their function w.r.t. the flow of pedestrians within the continuum model.

Model parameter	Function
$\beta_{\delta=d}$	Influence of the density gradient of the current pedestrian class
$\beta_{\delta \neq d}$	Influence of the density gradient of the other pedestrian class
β_g	Influence of the global route choice component
β_l	Influence of the local route choice component

Table 2

Tested parameter settings, where $\beta_g + \beta_l = 10$ and $\beta_{\delta \neq d} + \beta_{\delta=d} = 10$.

Model parameter	Range	Stepsize
$\beta_{\delta \neq d}$	5–10	0.5
$\beta_{\delta=d}$	0–5	0.5
β_g	5–10	0.5
β_l	0–5	0.5

Given that the final traffic state needs time to develop and all simulations need to be analysed at a similar moment in time, one can only compare traffic states after a warm-up period. The first pedestrians have easily reached the exit after 30 s due to the limited size of the infrastructure, thus after 90 s the traffic state should have stabilized. The traffic state at this specific moment of time serves as an input for the sensitivity analysis.

5. Presentation and discussion of the results

The results of the sensitivity analysis are discussed in this section. After a thorough examination of the sensitivity of the continuum model with respect to the individual flow situations in Sections 5.1–5.4, the combined results are discussed in Section 5.5. For each of the flow situations the sensitivity of the continuum model is analysed based on the simulated macroscopic traffic state and the emergence of the crowd movement phenomena mentioned in Section 4.1. Given that only one group of pedestrians is present in the simulation of uni-directional flow situations, the parameters $\beta_{\delta=d}$ and $\beta_{\delta \neq d}$ do not play a role. Therefore, in these specific paragraphs (5.1 and 5.4) only the influence of β_g and β_s is studied.

5.1. Uni-directional flow bottleneck

According to the literature, two classic crowd movement phenomena arise during bottleneck movements, namely anticipation [34] and dispersion [33]. The influence of the parameters on the first of these two phenomena, dispersion, has been visualized in Fig. 3. One can see that differences in the spatial spread the area in which pedestrians are present arise as a result of different parameter settings.

The figure reveals that for low values of β_g congestion arises near the walls of the bottleneck, while for relatively high values of β_g this is not the case since the pedestrians are better distributed over space. This trend can be explained as follows. The local route choice should theoretically point in the direction of decreasing density gradient. As a result, the flow of pedestrians is directed away from congested areas. Thus, the more influential the local route choice becomes in pedestrians operational movement decisions, the more equally the distribute over space.

Table 3 shows the width of the dispersion of the density at several cross-sections just upstream and downstream of the bottleneck. One can see that depending on the chosen balance between the global and local route choice parameters more or less lateral dispersion is found. That is to say, the flow of pedestrians becomes less and less focused in the last few metres upstream and downstream of the bottleneck if the value of β_l/β_g increases.

The results depicted in Fig. 3 and Table 3 show that the traffic state differs greatly depending on the influence of the local route choice. Given that several studies have indicated that bottleneck movements are funnel-shaped, it can be deduced that both relatively low ($\beta_l/\beta_g \leq 0.05$) and high ($\beta_l/\beta_g \geq 0.99$) values do not produce realistic predictions.

The influence of the parameters of the continuum model on the second crowd phenomena, namely the anticipation effect, have proven difficult to predict. Duives et al. [34] indicate the existence of anticipation, when a horizontal line at a non-zero velocity is present in the fundamental diagram. None of the resulting fundamental diagrams contain evidence that this line exists. Because the social force model assumes optimal operative walking behaviour, which causes pedestrians to sidestep rather than wait, this is a logical result. Consequently, it is concluded that the continuum model cannot predict anticipation behaviour upstream of a bottleneck.

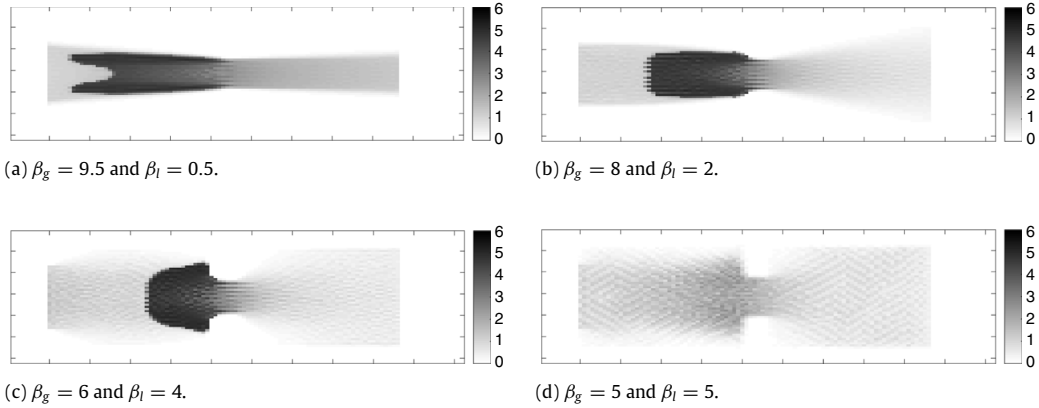
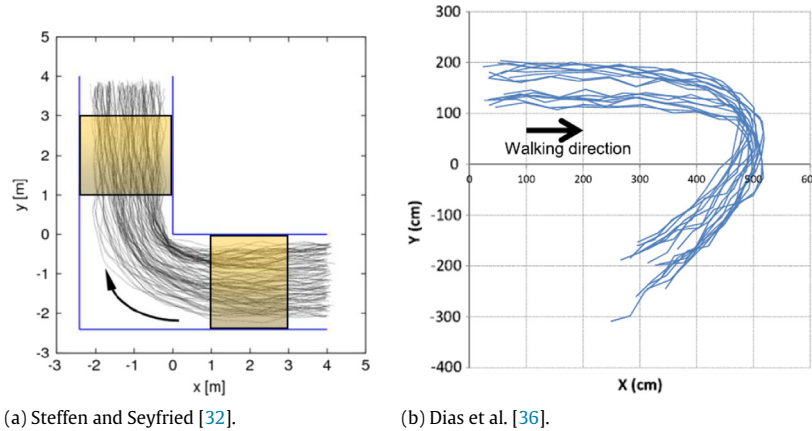
5.2. Uni-directional flow corner

Uni-directional flows around a 90° bend are often encountered. The research literature does, however, show that this flow situation only recently captured the attention of the pedestrian simulation community (among others Steffen and

Table 3

Width of the flow funnel upstream and downstream of the bottleneck, where w represents the distance with respect to the nearest wall of the bottleneck, $\beta_g = 10 - \beta_l$ and $\bar{q}_d = 0.25$ P/m/s.

			Local route choice— β_l								
			0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5
Demand (m)	Upstr.	$w = 1$ m	4.5	4.5	5	5	5.5	5.5	6	6	6.3
		$w = 2$ m	4.8	5.3	5.5	5.5	6.5	7	7.8	8.3	8.3
	Downstr.	$w = 1$ m	3	3.5	4	4	4.5	4.5	5	5.5	6
		$w = 2$ m	3.5	3.5	4	4.5	5	5.5	6.3	6.5	7.5

**Fig. 3.** Depictions of traffic state at a bottleneck for distinct combinations of β_g and β_l and a demand of $\bar{q}_d = 0.5$ P/m/s.**Fig. 4.** Depictions of trajectories captured in other empirical studies into walking behaviour around a 90° bend.

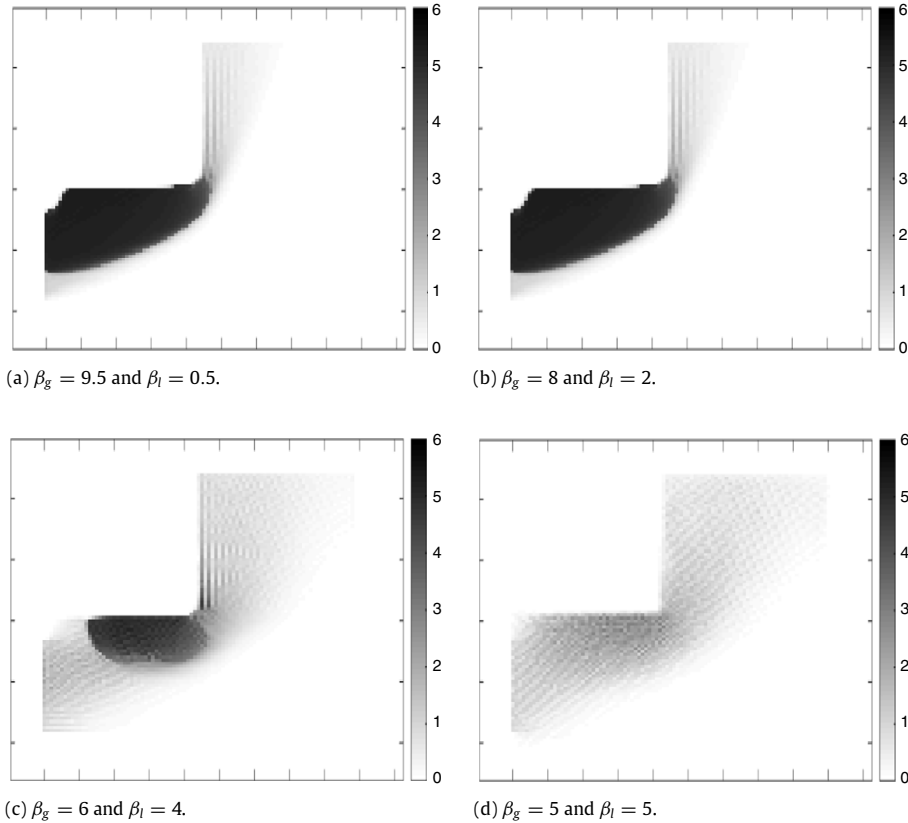
Seyfried [32] and Dias et al. [36]). Especially the effect of the turning angle on the flow rate has been studied. Steffen and Seyfried [32] and Dias et al. [36] both provided a visualization of the trajectories found during one of their experiments, see Fig. 4. In both studies only a slight narrowing of the flow is visible (i.e. the pedestrians move only lightly towards the inside of the bend while turning), which is quite unexpected since one would assume pedestrians to take the shortest route through the inside of the corner in cases of limited demand. Additionally, both studies do not record flow break down at low and medium densities.

Fig. 5 illustrates that local route choice has a large impact on the traffic state predicted by the continuum model after 90 s. Similar to the uni-direction bottleneck movements, relatively low values of β_l result in unrealistic predictions of congestion at the upstream inside of the corner, while relatively high values of β_l result in uncongested movements. The unrealistic predictions are due to the dominant influence of the global route choice, which represents the desire of pedestrians to take the shortest path through space. Thus, it can be concluded that the chosen parameter settings of the continuum model have a very large impact on the predicted efficiency of the pedestrian movements around a corner.

None of the traffic states visualized in Fig. 5 depict the expected low density zone downstream of the corner the inside wall. Since the global and local route choice both do not create a force which directs the flow away from the downstream

Table 4Width of the flow funnel in metres upstream and downstream of the bottleneck, where $\beta_g = 10 - \beta_l$.

			Local route choice— β_l									
			0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
Demand	$\bar{q}_d = 0.125$ P/m/s	1	1.8	2.0	2.0	2.0	2.3	2.5	2.5	2.5	2.5	2.5
		2	2.0	2.0	2.0	2.3	2.5	3	3.3	3.3	3.5	3.5
	$\bar{q}_d = 0.25$ P/m/s	1	2.5	2.5	2.5	2.8	2.8	3.5	4	5	5.3	5.5
		2	2.5	2.5	2.8	3.0	3.8	4.0	4.8	5.5	5.8	6.0

**Fig. 5.** Depictions of traffic state at the corner for distinct combinations of β_g and β_l and a demand of $\bar{q}_d = 0.25$ P/m/s.

inside wall, it is concluded that this phenomena cannot be simulated by means of the current version of the continuum model. It is expected that this crowd movement phenomena develops due to a force which is not yet taken into account in the continuum model, namely inertia.

In addition, the results show that in almost all cases the flow narrows considerably at the corner. Table 4 depicts the dispersion of pedestrians at a 45° angle, measured from the inside of the bend. Based on the literature one would expect the dispersion of the flow at the corner to approximate the width of the flow upstream and downstream of the corner. In this specific case study, this would result in a dispersion between 4 and 6 m. Given that a realistic prediction by means of the continuum model would need to resemble the trajectories shown in Fig. 4, it can be deduced that the continuum model only predicts the movements around a corner realistically if the fraction β_l/β_g is fairly high (i.e. $\beta_l/\beta_g \leq 0.8$).

5.3. Bi-directional flow straight

The development of lane formation, a well-known self-organization phenomena in bi-directional flows, is often used in the research literature to establish the validity of predictions generated by pedestrian simulation models. Hoogendoorn et al. [1] show that lane formation by means of the continuum model.

The simulation results, see Fig. 6 for some realizations, illustrate that the predicted traffic state differs greatly between the chosen parameter sets. For low fractions β_l/β_g and $\beta_{\delta=d}/\beta_{\delta \neq d}$ lane formation does develops briefly, yet the traffic flow stagnates completely within seconds of formation. It is generally assumed that lane formation counteracts deadlock. Yet, when the global route choice is very dominant and the lane formation is weakly defined (i.e. $\beta_{\delta=d}/\beta_{\delta \neq d}$ is small), the flows

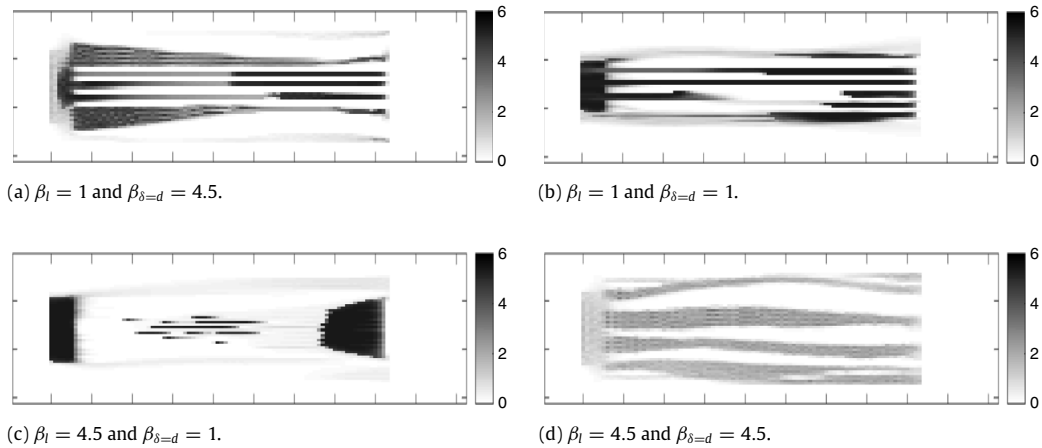


Fig. 6. Depictions of realized traffic state for several combinations of β_l/β_g and $\beta_{\delta=d}/\beta_{\delta\neq d}$ in a straight corridor under the influence of a bi-directional flow and a demand of $\vec{q}_d = 0.25$ P/m/s. $t = 90$ s.

Table 5

Nr. of lanes in the bi-directional flow, where the number accounts for the number of lanes, N = no self-organization visible, U = unstable flow system, $\beta_l = 10 - \beta_g$ and $\beta_{\delta\neq d} = 10 - \beta_{\delta=d}$.

		Influence other group						
		0.5	1.5	2.5	3.5	4.0	4.5	5.0
Local route choice	0.5	B	B	B	B	B	B	B
	1.5	B	B	B	B	B	B	N
	2.5	B	B	B	B	B	B	N
	3.5	B	B	B	B	B/L	B/L	N
	4.0	B	B	B	B/L	4	6	N
	4.5	B	B	8	6	5	7	N
	5.0	B	B/L	B/L	U	U	U	N

are not separated enough to limit the increase of the density and thus the decline of walking velocity that creates the on-set of congestion. Also low values of β_l/β_g and high values of $\beta_{\delta=d}/\beta_{\delta\neq d}$ have this effect. On the other hand, also high values of β_l/β_g do not automatically produce lane formation. Only for combinations of parameter settings in which β_l/β_g and $\beta_{\delta=d}/\beta_{\delta\neq d}$ are both relatively high, lane formation occurs.

Besides that, the results indicate that self-organization only takes place when the influence of the other group and the local route choice are both accounted for. Table 5 depicts the results of an analysis of the resulting traffic states. In the cases where “Blockage” is indicated, the dead lock occurs within the first 40 s of running the simulation. As one can see, only for values of $0.5 < \beta_{\delta=d}/\beta_{\delta\neq d} < 1$ lane formation does occur. Furthermore, the impact of $\beta_{d'\rightarrow d}$ on the development of lane formation seems to be also important, given that only for relatively high values ($0.8 \leq \beta_l/\beta_g < 1$) lane formation arises. As such, it can be concluded that only parameter settings which are in-between these two bounds produce realistic results. That is to say, realistic results are produced when the relative weight of the route choice components is $0.667 \leq \beta_l/\beta_g < 1$ and the relative weight of the different classes is $0.33 \leq \beta_{\delta=d}/\beta_{\delta\neq d} < 1$.

Besides the results with respect to the sensitivity of the traffic state, one other result needs to be noted. The continuum model only predicts the development of relatively small lanes. In practice, both small Hoogendoorn et al. [37] and very wide lanes Duives et al. [38] occur. It is unclear why the model predicts only one of these two possible solutions. It is hypothesized that this is due to the symmetric and stable design of the case study and the initialization process.

5.4. Intersecting flows 90 degrees

As mentioned before, the formation of stripes in situations where intersecting flow are present, which is sparingly recorded in the empirical research literature, can provide an indication of realistic simulation results. General descriptions of this flow situation in the literature do not mention densities at which flow breakdown occurs, though grid-lock is never mentioned for low densities and medium densities. Fig. 7 provides a visual overview of the traffic states and the self-

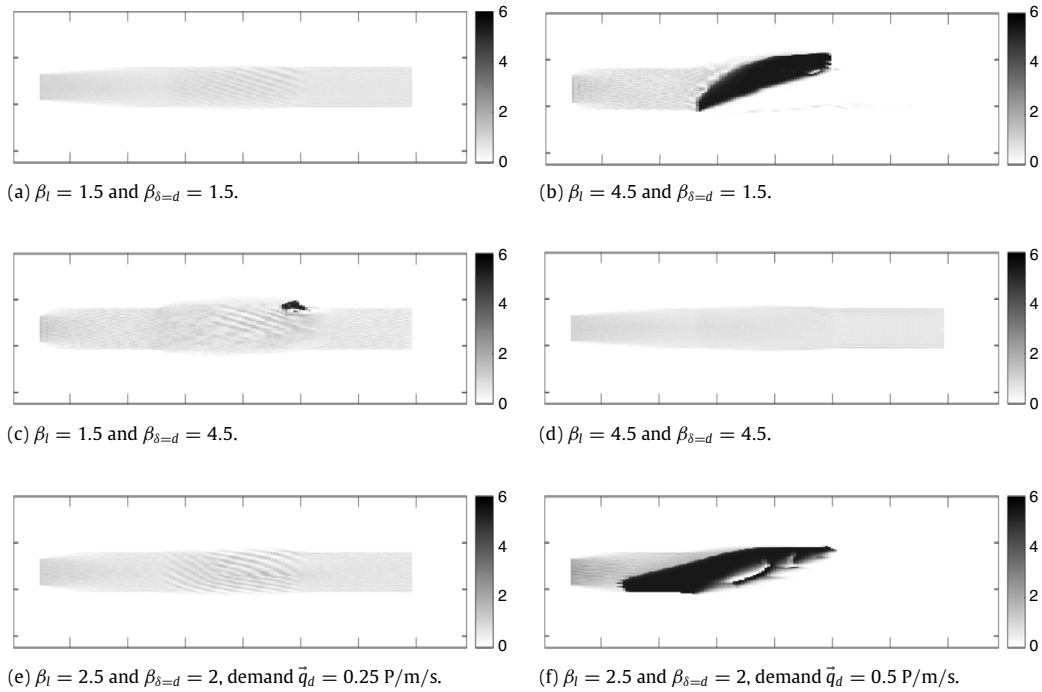


Fig. 7. Depictions of realized traffic state for several combinations of β_l/β_g and $\beta_{\delta=d}/\beta_{\delta \neq d}$ at an intersection.

Table 6

Nr. of lanes in the intersecting flow, where the number accounts for the number of lanes, N = no self-organization visible, B = blockage, U = unstable flow system, $\beta_g = 10 - \beta_l$ and $\beta_{\delta=d} = 10 - \beta_{\delta \neq d}$.

		Influence other group						
		0.5	1.5	2.5	3.5	4.0	4.5	5.0
Local route choice	0.5	24	20	18	13	11	U	U
	1.5	31	32	30	23	23	U	U
	2.5	32	33	26	24	24	U	U
	3.5	29	30	26	23	23	U	U
	4.0	B	B	22/B	23	U	U	-2
	4.5	B	B	B	22	U	-2	-2
	5.0	B	B	B	18	U	-4	-3

		Influence other group						
		0.5	1.5	2.5	3.5	4.0	4.5	5.0
Local route choice	0.5	B	B	B	B	B	B	B
	1.5	30	30	28	22	21	U	U
	2.5	30	30	26	23	24	U	U
	3.5	B	B	29	B/L	22	-4	-4
	4.0	B	B	B	B/L	23	-3	-4
	4.5	B	B	B	B/L	B/L	B/L	U
	5.0	B	B	B	U	U	U	U

organization patterns predicted by the continuum model. It can be seen that also in this case the parameter settings have a large impact on the traffic state. In some cases free flow arises, while in other cases total gridlock are predicted.

Table 6 presents an analysis of the traffic states after 90 s for different levels of demand. From the table it can be deduced that the development of stripe formations in the continuum model is dependent not only on the adopted parameter sets, but also on the demand at the intersection. If a low demand is present stripe formation arises for all parameter sets contained within the space $0.05 \leq \beta_l/\beta_g < 0.81$ and $0.33 \leq \beta_{\delta=d}/\beta_{\delta \neq d} < 1$, while for very high demands no stripe formation arises at all. At a medium demand level stripe formation does appear for parameter sets contained within the region of $0.18 \leq \beta_l/\beta_g < 0.81$ and $0.05 \leq \beta_{\delta=d}/\beta_{\delta \neq d} < 1$. This indicates that the local route choice behaviour is essential for stripe formation to develop. However, contrary to the expectation, it can also be concluded that predominantly a low influence of the local route choice on the overall movement decision of pedestrians gives rise to stripe formation. That is to say, the presence of local route choice behaviour makes the development of stripe formation possible since it forces the flows to cluster together per class. Yet, at the same time, a too dominant local route choice counteracts the development of stripe formation because too much clustering. Also the value of the fraction $\beta_{\delta=d}/\beta_{\delta \neq d}$ influences the development of stripe formation, though to a lesser extent. This is probably due to the fact that the influence of this fraction is partly regulated by the fraction β_l/β_g . As such, for higher values of β_l/β_g the effect of $\beta_{\delta=d}/\beta_{\delta \neq d}$ also increases.

Table 7

Summary of parameter settings at which flows can be modelled realistically by the Continuum model presented in Ref. [1].

	Phenomena	β_l/β_g	$\beta_{\delta=d}/\beta_{\delta\neq d}$
Bottleneck	Dispersion	$0.05 \leq \beta_l/\beta_g < 1$	n/a
	Anticipation	X	X
Corner	Congestion upstream of bottleneck	$\beta_l/\beta_g \geq 0.11$	n/a
	Low density downstream bottleneck inside corner	X	X
	Dispersion ≤ 6 m	X	X
Bi-directional	Lane-formation	$0.67 \leq \beta_l/\beta_g < 1$ $\bar{q}_d = 0.25$ P/m/s:	$0.33 \leq \beta_{\delta=d}/\beta_{\delta\neq d} < 1$ $\bar{q}_d = 0.25$ P/m/s:
Intersecting	Stripe formation	$0.05 \leq \beta_l/\beta_g < 1$	$0.33 \leq \beta_{\delta=d}/\beta_{\delta\neq d} < 1$
		$\bar{q}_d = 0.5$ P/m/s:	$\bar{q}_d = 0.5$ P/m/s:
		$0.018 \leq \beta_l/\beta_g < 0.81$	$0.05 \leq \beta_{\delta=d}/\beta_{\delta\neq d} < 1$
Inner bounds		$0.66 \leq \beta_l/\beta_g < 0.81$	$0.33 \leq \beta_{\delta=d}/\beta_{\delta\neq d} < 0.81$

5.5. Combining the results of the sensitivity analysis

Within this study several limitations have been established with respect to the choice of parameters of the continuum model in order to predict crowd movement dynamics realistically. In Table 7 these bounds have been summarized. One can see that especially the desire to predict the development of lane formation in the bi-directional flow situation and stripe formation in the intersecting flow situation imposes stringent bounds on the parameters β_l and β_g . This desire also bounds the parameters which describe the influence of the density gradient of the own and other group. The uni-directional flow scenarios only provide bounds on the weighting of both route choice components.

Given that one is searching for a parameter set which can simulate the movements of pedestrians realistically in all four flow situations, all bounds mentioned in Table 7 need to be adhered to. That is to say, parameter sets which adhere to these bounds ensure that the prediction of the crowd movements mimics the phenomena mentioned in the research literature. Consequently, it is concluded that several parameter sets exist which produce a realistic prediction of the crowd movement dynamics mentioned in the literature, since several parameter sets reside in the parameter space bounded by $0.66 \leq \beta_l/\beta_g < 0.81$ and $0.33 \leq \beta_{\delta=d}/\beta_{\delta\neq d} < 0.81$.

This does, however, not directly imply that all parameter sets within these bounds provide quantitatively correct predictions. More research is necessary to test the quantitative validity of the continuum model for the parameter sets that lie within the area specified by the limitations on the fractions β_l/β_g and $\beta_{\delta=d}/\beta_{\delta\neq d}$.

6. Conclusions and future research

This paper has provided a comprehensive sensitivity analysis of the parameters of the continuum model proposed by Hoogendoorn et al. [1]. A first estimation of parameter sets which provide realistic results is performed based on the qualitative comparison between the simulation results of the continuum model and the behavioural trends described in the research literature. By means of four distinct case studies (i.e. a uni-directional bottleneck, a uni-directional corner, a bi-directional straight and a intersecting flow situation) the sensitivity of continuum model's prediction with respect to the parameters that govern the simulation of the operational walking behaviour within the model were put to the test.

It was found that most self-organization phenomena and behavioural trends only develop under very specific conditions, and as such can only be simulated using specific parameter sets. Only parameter sets which lie within the parameter space bounded by $0.66 \leq \beta_l/\beta_g < 0.81$ and $0.33 \leq \beta_{\delta=d}/\beta_{\delta\neq d} < 0.81$ provide realistic results, given that one is searching for a parameter set which can simulate the flow situations tested in this paper realistically. Based on these results, it is furthermore concluded that the incorporation of local route choice behaviour and the balancing of the aptitude of pedestrians with respect to their own class and other classes are both essential in the correct prediction of crowd movement dynamics. The exact parameter set which provides the best prediction within this region cannot be established based on the qualitative assessment performed in this study. Calibration based on a quantitative comparison of realized and predicted traffic states is necessary to establish the optimal parameter set(s).

This study has provided a first step in the calibration and validation of the continuum model. Future work will attempt to calibrate and validate the model quantitatively using empirical data sets. Additionally, the sensitivity analysis uncovered two flaws in the current functioning of the model. The current version of the continuum model is not capable of simulating anticipation and the low density zone downstream of the corner. Attempts will be made to alleviate these two issues.

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